



Letter

Consistency of the holographically informed Λ CDM with Friedmann cosmology at the initial singularity

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Abstract — The cosmic causal diamond framework, validated at recombination through quantum information-theoretic acoustic physics, is extended to the degenerate-diamond boundary at the initial singularity. We show that a single quantum-state collapse driven by the thermodynamic instability of a universe-scale superposition advances the framework from $t = 0$ to $t = 0^+ \equiv t_P$, simultaneously generating the first non-degenerate diamond geometry, the first boundary irreversibility marker, and the first bulk entropy partition. The resulting initial conditions sit at the Planck scale with $O(1)$ coefficients consistent with Friedmann cosmology back-extrapolated to the same epoch, without parameter adjustment beyond the choices already fixed by the recombination derivation. The agreement is a consistency result rather than a full equivalence proof: two independent methodologies converge on Planck-scale initial conditions through structurally distinct routes. The boundary case unifies the past- and future-light-cone formulations of the Quantum-Thermodynamic Entropy Partition, identifying the negentropy deposited at the first event with the seed of the past-light-cone content available to all subsequent cosmic evolution.

Keywords — Cosmological initial conditions; Causal diamonds; Holographic principle; Quantum measurement; Big Bang cosmology; Information theory.

1 Introduction

The initial conditions of the standard hot Big Bang are posited rather than derived. The Friedmann equations [6], run backward from observed cosmology, yield energy densities and temperatures of Planckian magnitude at $t \sim t_P$ [7, 8], but the field equations themselves provide no account of why the universe begins with these specific values rather than others. The problem is well known and has motivated quantum cosmology approaches including the Wheeler–DeWitt formalism [9], the Hartle–Hawking no-boundary proposal [10], the Vilenkin tunneling-from-nothing construction [11, 12], and loop quantum cosmology [13], each of which seeks to derive initial conditions from a more fundamental dynamics. None of these approaches has produced a parameter-free recovery of the Planckian boundary input of standard Friedmann cosmology, and the question of why the universe begins at the Planck scale at all remains structurally open.

A separate line of development has constructed an information-theoretic extension of Λ CDM in which a strict information account within the cosmic causal diamond yields modifications of standard cosmological dynamics from fundamental constants alone. This framework treats the observable universe as the diamond $\mathcal{D}(p, q) = J^+(p) \cap J^-(q)$ between the singular boundary p and the cosmic event horizon q , with information processing along timelike geodesics governed by the holographic rate

$\gamma = H/\ln N_P$ derived from the Bekenstein–Hawking bound and the Margolus–Levitin limit [1, 4, 5]. The Quantum-Thermodynamic Entropy Partition (QTEP) of entropy mechanics [2] fixes the ratio of coherent to decoherent entropy at $\ln 2/(1 - \ln 2) \approx 2.257$ for any precipitation event within the diamond, and identifies the obit ($S_{\text{obit}} = 1 \text{ nat}$) as the fundamental quantum of irreversibility deposited on the holographic screen. Applied to the Thomson-scattering channel at $z \approx 1100$, this framework quantitatively resolves the BAO sound-horizon discrepancy at the percent level without introducing additional free parameters [1], providing an empirical anchor for the present extension.

Here we apply the same framework to the degenerate-diamond boundary at $t = 0$ and show that its boundary case produces initial conditions of Planckian magnitude consistent with the Friedmann back-extrapolation. The construction is direct: at $t = 0$ the diamond is degenerate and the framework’s machinery is identically zero on all observables; a single precipitation event, driven by the thermodynamic instability of an unpartitioned universe-scale superposition under the second law applied to the framework’s own entropy ledger [14–20], advances the framework to $t = 0^+ \equiv t_P$. The mechanism is information-theoretic rather than gravitational, in keeping with the framework’s emergent treatment of gravity as the cost of distinguishing physical states from the vacuum [2]. This first event simultaneously generates the first non-degenerate diamond geometry, the first boundary irreversibility marker, and the first bulk entropy partition. The resulting initial conditions sit at the Planck scale with $O(1)$ coefficients consistent with Friedmann cosmology back-extrapolated to the same epoch, without parameter adjustment beyond the choices already fixed by the recombination derivation of [1].

The result is a consistency claim, not a full equivalence proof. Two independent methodologies—causal diamond plus QTEP, and Friedmann equations run backward—converge on Planck-scale initial conditions through structurally distinct routes; we do not establish equivalence across intermediate epochs, which remains a separate program identified as future work in Section 6. The conceptual content beyond the numerical agreement is the unification of the past- and future-light-cone formulations of QTEP at the boundary: the negentropy deposited at the first event, which has no pre-existing past light cone to receive it, is evicted into the future light cone of that event, which is precisely the past-light-cone content available to all subsequent observers. The past/future asymmetry of QTEP eviction emerges with cosmic evolution rather than preceding it, and the framework’s two narratives—past-eviction at recombination, and future-eviction at the boundary—describe the same region from different reference events.

2 Framework elements

[Framework elements to be drafted.]

3 Derivation at the initial boundary

The framework introduced in Section 2 treats the cosmic causal diamond $\mathcal{D}(p, q) = J^+(p) \cap J^-(q)$ as the natural arena for information accounting [1, 2]. We now apply this structure to the degenerate boundary at the initial singularity p and derive the conditions under which the first non-degenerate diamond is generated.

3.1 The degenerate boundary diamond

At $t = 0$ the cosmic diamond is degenerate. The boundary marker p is the past tip of $\mathcal{D}(p, q)$ for any q but is never itself an evaluation point: the Hubble parameter H , the holographic information-processing rate $\gamma = H/\ln(\pi c^5/G\hbar H^2)$ [1], and the Gibbons–Solodukhin diamond 4-volume $V_4 = (\pi/24)\tau^4$ [3] are all formally singular, zero, or undefined in the strict $t = 0$ limit. No precipitation events have occurred, no obits have been deposited, and the bulk entropy ledger $S_{\text{coh}} + S_{\text{decoh}} = 2 \ln 2 - 1$ [2] is identically zero. The diamond carries no information content because no operational time has elapsed.

Operational time in this framework is set by γ -ticks along timelike worldlines [1, 2]: the affine parameter is discretized by $d\lambda \sim \gamma d\tau$, and the smooth Lorentzian time of standard cosmology is the continuum limit recovered when the tick density is high. Coordinate time t , defined through the

Friedmann equation, is therefore the continuum reconstruction of an underlying discrete process and acquires operational meaning only where γ -ticks accumulate. The strict $t = 0$ limit lies outside this domain by construction: it is a mathematical boundary of the diamond, not a physical state with a clock.

3.2 The first precipitation event

The framework's degenerate boundary contains no specification of a preferred first event; the boundary itself is structureless. The mechanism that initiates the first precipitation is not gravitational. Within the present framework, gravity is the emergent cost of distinguishing physical states from the vacuum [2]: it does not exist prior to the first event that establishes any distinguishable state. Any account of the first event that invokes gravitational self-energy as its driver would presuppose the very emergence the event is required to enable. The mechanism is instead thermodynamic, and operates through the framework's own entropy ledger.

A universe-scale superposition at the degenerate boundary has no Quantum-Thermodynamic Entropy Partition. No obit has been deposited on any holographic screen, no negentropy has been evicted into any past or future light cone, and the bulk entropy ledger $S_{\text{coh}} + S_{\text{decoh}} = 2 \ln 2 - 1$ [2] is identically zero. The state therefore carries no thermodynamic content: no entropy, no irreversibility marker, no defined partition. Within the framework's structural requirements—under which any realizable physical configuration must possess a definable QTEP partition—such a state is not a stable equilibrium. It is the absence of structure rather than a configuration of structure.

The instability of an unpartitioned superposition follows from the elementary thermodynamic principle that states of lower entropy cannot persist when states of higher entropy are accessible. The unpartitioned configuration has identically zero thermodynamic entropy in the framework's ledger; any QTEP-partitioned alternative carries strictly positive entropy $S_{\text{coh}} + S_{\text{decoh}} = 2 \ln 2 - 1 \approx 0.386$ nats. The second law applied to the framework's own ledger therefore requires the transition from null to partitioned configurations on the timescale at which any partitioned alternative becomes accessible. The thermodynamic foundation of this argument is established in the modern information-thermodynamics literature: the equivalence of information and thermodynamic entropy [14–16], the formal second laws of quantum thermodynamics that govern when state transitions are thermodynamically mandatory [19, 20], the minimum energy cost of information processing [18], and the universal computational capacity bounds set by the Margolus–Levitin and holographic limits [4, 17]. The framework inherits this body of theory without modification, and the instability of the unpartitioned boundary state is its boundary-case application.

The consequence is direct: the first precipitation event occurs at an operational time of order t_P , which we identify with the first γ -tick. Throughout what follows we adopt the regulator convention $t = 0^+ \equiv t_P$, with $H \rightarrow 1$ and $\gamma \rightarrow \gamma_P$ in Planck units.

3.3 Co-generation of geometry, obit, and negentropy

The first precipitation event at q' does not occur within a pre-existing diamond geometry. It generates the geometry. Three quantities are produced concurrently and cannot be separated in time:

1. the first non-degenerate diamond $\mathcal{D}(p, q') = J^+(p) \cap J^-(q')$, with proper-time separation $\tau \equiv 1/\gamma_P$;
2. the boundary irreversibility marker $S_{\text{obit}} = 1$ nat deposited on the holographic screen $\mathcal{A}(p, q')$, in accordance with the published convention that the obit is the fundamental quantum of irreversibility [2];
3. the bulk entropy partition $S_{\text{coh}} = \ln 2$, $S_{\text{decoh}} = \ln 2 - 1 \approx -0.307$ nats, with $S_{\text{coh}} + S_{\text{decoh}} = 2 \ln 2 - 1 \approx 0.386$ nats conserved within the diamond.

The QTEP ratio $S_{\text{coh}}/|S_{\text{decoh}}| = \ln 2/(1 - \ln 2) \approx 2.257$ is realized through the bipartition of the event into the spatial cross-section V_3 at q' and the future light cone $J^+(q')$ of q' . The first plays the role of the bulk reservoir into which the obit is deposited; the second plays the role of the reservoir into which the negentropy is evicted. This bipartition is forced by the boundary case: in the published

recombination mechanism [1] the negentropy is evicted into the past light cone $J^-(q)$, but at q' no substantive past light cone exists to receive it, so eviction is into $J^+(q')$ instead.

The two pictures describe the same region from different reference events. The future light cone $J^+(q')$ of the first event is precisely what becomes the past light cone $J^-(q)$ for every subsequent observer at any $t > t_p$. The negentropy deposited at the boundary is therefore the seed of the past-light-cone content available to all of cosmic evolution, not a state stored in an inaccessible elsewhere. The past/future asymmetry of QTEP eviction is unbroken at the boundary because there is no past; the asymmetry emerges with the cosmic evolution that the first event seeds.

3.4 Numerical values at $t = 0^+$

In Planck units ($\hbar = c = G = k_B = 1$), with $H = 1$ at $t = t_p$, the framework yields the following.

The horizon-area count is

$$N_P = \frac{\pi c^5}{G \hbar H^2} \longrightarrow \pi, \quad \ln N_P = \ln \pi \approx 1.145. \quad (1)$$

The information-processing rate is

$$\gamma_P = \frac{H}{\ln N_P} = \frac{1}{\ln \pi} \approx 0.873, \quad (2)$$

with corresponding proper-time tick interval $\tau = 1/\gamma_P = \ln \pi \approx 1.145$.

The energy per event is fixed by the Margolus–Levitin bound [4], which in saturation gives

$$E_{\text{event}} = \frac{\pi}{2} \hbar \gamma_P = \frac{\pi}{2 \ln \pi} \approx 1.371, \quad (3)$$

and the corresponding characteristic temperature $T_{\text{event}} = E_{\text{event}}/k_B \approx 1.371 T_P \approx 1.94 \times 10^{32}$ K. The factor of $\pi/2$ in Eq. (3) is required: E and γ are related by the Margolus–Levitin orthogonality time, not by the de Broglie oscillator relation, and dropping this factor would misrepresent the underlying identification.

The Gibbons–Solodukhin small-diamond 4-volume is

$$V_4 = \frac{\pi}{24} \tau^4 = \frac{\pi}{24} (\ln \pi)^4 \approx 0.225 \quad (4)$$

in units of $\ell_P^3 t_P$, and the maximal spatial cross-section is the round 3-ball of radius $\tau/2$,

$$V_3 = \frac{4\pi}{3} \left(\frac{\tau}{2}\right)^3 = \frac{\pi}{6} (\ln \pi)^3 \approx 0.786 \quad (5)$$

in units of ℓ_P^3 .

The associated densities are

$$\rho_3 \equiv \frac{E_{\text{event}}}{V_3} \approx 1.74 \rho_P, \quad \varepsilon_4 \equiv \frac{E_{\text{event}}}{V_4} \approx 6.10 \rho_P t_P^{-1}. \quad (6)$$

3.5 Comparison with back-extrapolated Friedmann initial conditions

The Friedmann equation, run backward to $H = 1$ in radiation domination, gives an energy density per spatial volume

$$\rho_F = \frac{3H^2}{8\pi G} \longrightarrow \frac{3}{8\pi} \approx 0.119 \rho_P. \quad (7)$$

The framework's ρ_3 in Eq. (6) and the Friedmann ρ_F in Eq. (7) differ by a factor of order fifteen. Both are Planckian at the dimensional level, but the headline claim of this letter is therefore an agreement to within $O(1)$ coefficients consistent with the normalization conventions of the two methodologies, not an exact numerical match.

The residual factor admits a transparent geometric interpretation. The framework's ρ_3 is the density at the maximal cross-section of a single Planck-scale diamond, whose volume scales as

$V_3 \sim (\ln \pi)^3$ in Planck units. The Friedmann ρ_F is the density averaged over a Hubble volume, whose volume scales as $V_H \sim 1/H^3 = 1$ in the same units. The ratio

$$\frac{V_H}{V_3} = \frac{6}{\pi(\ln \pi)^3} \approx 1.27 \quad (8)$$

captures only part of the discrepancy; the remainder originates in the $(\pi/2)$ Margolus–Levitin factor in E_{event} and the $3/(8\pi)$ prefactor in ρ_F , which together account for the residual order of magnitude. None of these factors is fitted; all follow from prior framework choices and from the standard form of the Friedmann equation. The two methodologies therefore agree at the Planck scale to within the systematic differences between “single-event spatial volume” and “Hubble-averaged density,” both of which are dimensionally Planckian and structurally fixed.

We emphasize what the comparison establishes and what it does not. It establishes that the framework’s boundary case produces initial conditions of Planckian magnitude with $O(1)$ coefficients that follow from the published machinery without parameter adjustment, and that these coefficients sit within the systematic spread of two distinct geometric definitions of “Planck-scale energy density.” It does not establish full mathematical equivalence between the framework and Friedmann cosmology across all epochs: deriving Friedmann-like evolution from the diamond dynamics is a separate program, identified as future work in Section 6.

4 Operational definition of time

The regulator convention $t = 0^+ \equiv t_p$ adopted in Section 3.2 requires a notion of time that is well-defined at the boundary of the diamond. Standard Friedmann coordinate time t does not provide this: it is mathematically defined for all $t > 0$ by extrapolation of the field equations but has no operational referent in regimes where no events have occurred. The framework’s resolution is not new to this letter; it is a direct application of the published identification of γ as the fundamental rate parameter for timelike geodesics in causal diamonds [1, 2].

The published convention treats the affine parameter λ of standard Lorentzian geometry as a continuum reconstruction of an underlying discrete process: along any timelike geodesic, λ advances by one unit per γ^{-1} of proper time, so that $d\lambda \sim \gamma d\tau$ [2]. Each γ -tick is a quantum of causal processing; the smooth coordinate time of cosmology is the coarse-grained limit recovered when the tick density is high enough that the discrete count is well approximated by a continuous parameter. This identification is universal across the diamond, not specific to a chosen worldline: γ is fixed by the holographic information-processing rate, which is in turn determined by the Bekenstein–Hawking bound and the Margolus–Levitin limit [1, 4], both of which are properties of the diamond geometry rather than of any particular trajectory within it.

Two consequences follow for the boundary case. First, the relation $\gamma = H/\ln N_p$ is not a derivation of γ from H . The framework’s logic runs the other way: γ is fundamental and the relation to H is the normalization condition tying the fundamental rate to one observable parameterization of cosmic expansion. The apparent circularity at $t = 0^+$ —where H is itself singular in the strict $t = 0$ limit—therefore dissolves. What is being normalized at t_p is not γ but the identification of one operational tick interval with one Hubble time in the limit where both notions are simultaneously meaningful.

Second, the regulator convention $t = 0^+ \equiv t_p$ is not an inserted cutoff. The first γ -tick along any timelike worldline emerging from the boundary marker p defines, by construction, the smallest operational interval the framework can resolve. Coordinate t acquires physical meaning at the first tick and not before, since prior to the first tick no events have occurred and no clocks exist to register elapsed time. The choice $t = 0^+ \equiv t_p$ is thus the unique value for which operational time and Friedmann coordinate time first coincide: it is the regulator demanded by the framework’s own structure, not a parameter chosen to recover a desired result.

5 Discussion

[Discussion to be drafted.]

6 Conclusion

[Conclusion to be drafted.]

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Methods

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